

互联网协议实验

- 搭建实验环境

- **\$** sudo apt install wireshark ifupdown
- **\$** sudo mn --nat **#** allows hosts to connect to the Internet
- **mininet>** xterm h1
- **h1 #** echo "nameserver 8.8.8.8" > /etc/resolv.conf
- h1 # wireshark &

- 实验步骤

- h1 # wget -U "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/123.0 Safari/537.36" https://www.ucas.ac.cn

互联网协议实验

- 观察wireshark输出（一），以www.baidu.com页面为例
 - ARP协议: IP地址->MAC地址映射
 - DNS协议: 域名->IP地址映射
 - TCP协议: 数据传输

No.	Time	Source	Destination	Protocol	Length	Info
4	9.256069038	82:24:2a:96:c2:c8	Broadcast	ARP	42	Who has 10.0.0.3? Tell 10.0.0.1
5	9.259799814	72:e0:11:58:ce:4f	82:24:2a:96:c2:c8	ARP	42	10.0.0.3 is at 72:e0:11:58:ce:4f
6	9.259812291	10.0.0.1	159.226.39.1	DNS	73	Standard query 0x0393 A www.baidu.com
7	9.259815360	10.0.0.1	159.226.39.1	DNS	73	Standard query 0xd6fe AAAA www.baidu.com
8	9.308805004	159.226.39.1	10.0.0.1	DNS	132	Standard query response 0x0393 A www.baidu.com CNAME w
9	9.318736091	159.226.39.1	10.0.0.1	DNS	157	Standard query response 0xd6fe AAAA www.baidu.com CNAM
10	9.319227342	10.0.0.1	119.75.216.20	TCP	74	48488 → 80 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_P
11	9.338055235	119.75.216.20	10.0.0.1	TCP	58	80 → 48488 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=
12	9.338087433	10.0.0.1	119.75.216.20	TCP	54	48488 → 80 [ACK] Seq=1 Ack=1 Win=29200 Len=0
13	9.341933425	10.0.0.1	119.75.216.20	HTTP	194	GET / HTTP/1.1
14	9.342532422	119.75.216.20	10.0.0.1	TCP	54	80 → 48488 [ACK] Seq=1 Ack=141 Win=65535 Len=0
15	9.353627861	119.75.216.20	10.0.0.1	TCP	454	[TCP segment of a reassembled PDU]
16	9.353647250	10.0.0.1	119.75.216.20	TCP	54	48488 → 80 [ACK] Seq=141 Ack=401 Win=30016 Len=0
17	9.362959249	119.75.216.20	10.0.0.1	TCP	1474	[TCP segment of a reassembled PDU]
18	9.362983464	10.0.0.1	119.75.216.20	TCP	54	48488 → 80 [ACK] Seq=141 Ack=1821 Win=32660 Len=0
19	9.363141969	119.75.216.20	10.0.0.1	HTTP	1015	HTTP/1.1 200 OK (text/html)
20	9.363161257	10.0.0.1	119.75.216.20	TCP	54	48488 → 80 [ACK] Seq=141 Ack=2782 Win=35500 Len=0
21	9.363337625	119.75.216.20	10.0.0.1	TCP	54	80 → 48488 [FIN, ACK] Seq=2782 Ack=141 Win=65535 Len=0
22	9.366398955	10.0.0.1	119.75.216.20	TCP	54	48488 → 80 [FIN, ACK] Seq=141 Ack=2783 Win=35500 Len=0
23	9.367574703	119.75.216.20	10.0.0.1	TCP	54	80 → 48488 [ACK] Seq=2783 Ack=142 Win=65535 Len=0

互联网协议实验

- 观察Wireshark输出 (二)
 - 不同层次的协议封装
- Ethernet < IP < UDP < DNS
- Ethernet < IP < TCP < HTTP

```
26:fc:8f:07:34:76 (26:fc:8f:07:34:76), Dst: 6a:3a:d5:99:4a:0c (6a:3a:d5:99:4a:0c)  
Internet Protocol Version 4, Src: 172.0.0.1 (172.0.0.1), Dst: 61.135.169.121 (61.135.169.121)  
User Datagram Protocol, Src Port: 56443 (56443), Dst Port: http (80), Seq: 1, Ack: 1, Len: 111  
Protocol
```

```
(linux-gnu)\r\n
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\n  
e\r\n
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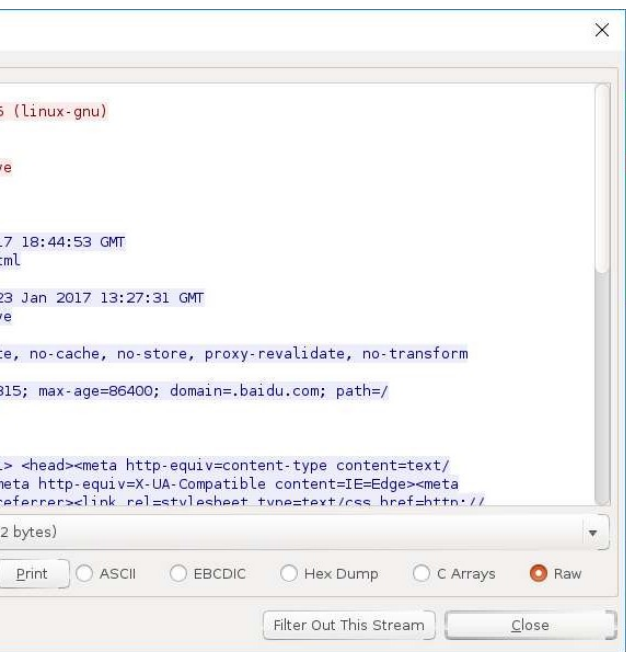
```
http://www.baidu.com/1
```

```
01
```

```
► Ethernet II, Src: 26:fc:8f:07:34:76 (26:fc:8f:07:34:76), Dst: 6a:3a:d5:99:4a:0c (6a:3a:d5:99:4a:0c)  
► Internet Protocol Version 4, Src: 172.0.0.1 (172.0.0.1), Dst: 1.2.4.8 (1.2.4.8)  
► User Datagram Protocol, Src Port: 58330 (58330), Dst Port: domain (53)  
▼ Domain Name System (query)  
    [Response In: 3]  
    Transaction ID: 0x7027  
    ► Flags: 0x0100 Standard query  
        Questions: 1  
        Answer RRs: 0  
        Authority RRs: 0  
        Additional RRs: 0  
    ▼ Queries  
        ► www.baidu.com: type A, class IN
```

互联网协议实验

- 观察Wireshark输出 (三)
 - TCP承载HTTP协议

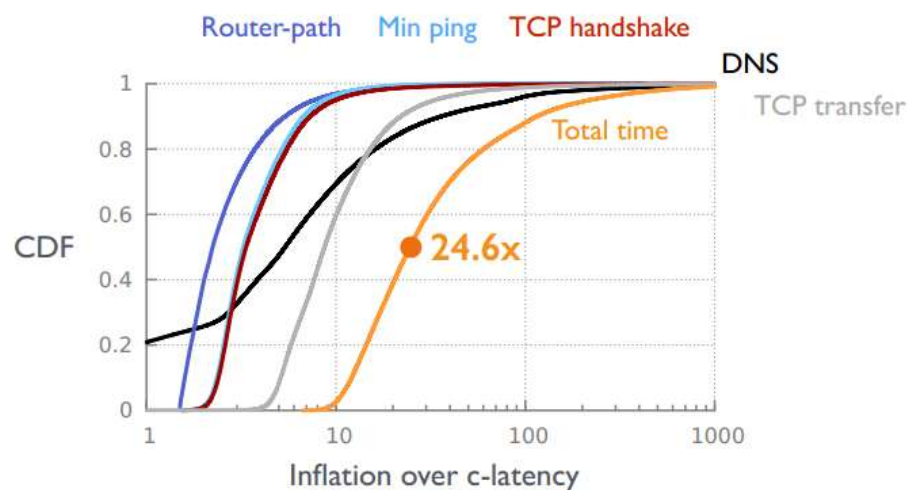


172.0.0.1	61.135.169.121	TCP	54	56443 > http [ACK] Seq=1 Ack=1 Win=29200 Len=0
172.0.0.1	61.135.169.121	HTTP	165	GET / HTTP/1.1
61.135.169.121	172.0.0.1	TCP	54	http >
61.135.169.121	172.0.0.1	HTTP	2835	HTTP/1.
172.0.0.1	61.135.169.121	TCP	54	56443 >
61.135.169.121	172.0.0.1	TCP	54	http >
172.0.0.1	61.135.169.121	TCP	54	56443 >
61.135.169.121	172.0.0.1	TCP	54	http >
6a:3a:d5:99:4a:0c	26:fc:8f:07:34:76	ARP	42	who has
26:fc:8f:07:34:76	6a:3a:d5:99:4a:0c	ARP	42	172.0.0
0.0.0.0	255.255.255.255	DHCP	342	DHCP Di
0.0.0.0	255.255.255.255	DHCP	342	DHCP Di
0.0.0.0	255.255.255.255	DHCP	342	DHCP Di
0.0.0.0	255.255.255.255	DHCP	342	DHCP Di
172.0.0.3	224.0.0.251	MDNS	87	Standar

on wire (1320 bits), 165 bytes captured (1320 bits) on interface 0
26:fc:8f:07:34:76 (26:fc:8f:07:34:76), Dst: 6a:3a:d5:99:4a:0c (6a:3a:)
version 4, Src: 172.0.0.1 (172.0.0.1), Dst: 61.135.169.121 (61.135.169)
il Protocol, Src Port: 56443 (56443), Dst Port: http (80), Seq: 1, Ack

传输性能测量实验

- 在Linux系统或Mininet环境下，编写Python测量脚本，复现如下实验结果
 - 只需要Ping、DNS、TCP transfer、Total time这几个值
 - 获取1000个左右网站的上述指标
 - 使用GeoIP数据库查询IP地址对应的地理位置，大致估算c-latency



实验内容

- 互联网协议实验

- 在节点h1上开启wireshark抓包，用wget下载www.ucas.ac.cn页面
- 调研说明wireshark抓到的几种协议
 - ARP, DNS, TCP, HTTP, HTTPS
- 调研解释h1下载ucas页面的整个过程
 - 几种协议的运行机制

- 传输性能测量实验

- 如果在Mininet环境下测量，`$ sudo mn --nat`
- 测量结果可能与原图不太一样：当前大部分通过CDN技术分发，服务器地址大都在北京
- 结合本讲的内容，调研解释这幅图中结果的成因